Script Grammar
Find out more about the script grammar in Hindi, and all the rules of stringing together letters **https://dgit.in/sep19-43**

Swalekh Flip
Download the Swalekh Flip keyboard which has unique features designed for easy editing **https://dgit.in/sep19-44**

research to fit in all the characters in a way that makes sense to native speakers. Considering there isn't a lot of information for the Indian languages available online, we analysed text corpora to figure out which characters to include and to determine how frequently they're used. Depending on the language, we tailored aspects of the layout, like the set of digits—for example, while English uses 0123456789, Hindi and other Indian languages written in Devanagari use ०१२३४५६७८९. Moreover, once we've built support for a language, we always invite a group of native speakers to test and fill out a survey to understand their typing experience."

Gboard also allows transliteration, which means that you can use English to key in the text in the Indic language. The swiping input combined with transliteration allows users to type rapidly in native languages, using a familiar QWERTY keyboard layout. At times there are multiple layout options available for particular language, which can be customised by the user. These include compact layouts which are smaller versions aimed at casual users. Daan says, "when we implement Gboard's layouts for a given language, sometimes we come across situations where multiple keyboard layouts are commonly used for that language. In that case, we sometimes choose to include multiple layouts for a given language, so each user can use the layout they prefer. That's also what happened with the compact layouts we offer for certain Indian languages." The users can also adjust the size of the keyboard, and change its position. The keyboard is not pinned to the bottom of the screen, and can be moved around anywhere, as per the need.

Reverie has a keypad for smartphones called Swalekh Flip. Pani says, "when we designed for Reverie, we were the first keypad to keep ten keys in one row. If you look at Swalekh, it has exactly ten keys in one row, and it also has only three rows. The idea is that the typing experience for any Indian language will be no poorer than what it can be for English."

One of the special properties of Indian languages, is that they have what is called script grammar. The way the letters form have rules, and this can be incorporated into the keypads to improve the typing experience. For example, the letter ङ (xn), can never be combined with a matra. There is still a provision for it within Unicode, and it is possible to type ङा (xna), which is never used. Pani says, "What we did is that we used the script grammar for Indian languages. Indian languages have a very well-defined and understood script grammar. Therefore, the keypad became much smarter and easier to use. Prediction and machine learning also there, but they are not rules. They are not your language property. They will help you to type fast. They will probably also help you to give you suggestions tuned to your usage behavior and so on. But it may not make the fundamental typing easy enough. To make the typing itself easier, you have to come up with the basic keypad layout and typing process." Now, the approach of keyboards allowsall kinds of nonsense formations that can never be actually used. For example, here is a whole bunch of matras piled onto a single प(pa): पौ्. Swalekh Flip prevents such nonsense formations, by using what they call a layering system. Pani explains, "if you have a keypad and you have a cursor, then you are in a base layer, where you can only type in a vowel or a consonant. If you are writing किताब (kitab), that means you write क (ka) which is a letter first, and then only you can add a matra (ि). When people designed typewriters or early printing typesets, because it was a physical need, they were forced to first place the matra when it came to the left. Whereas when you learn to write, you add the matra to the consonant. So the expectation from the keypad will also be like that. The second layer of Swalekh offers you that. After you have typed the consonant, only then the vowels also give you the option of typing the matra. Until that time, you cannot type the matra. If you have typed a matra, then the matra association with the consonant is over. Therefore there is no way you can actually feed in multiple matras."

Then comes the handling of the yuktakshars. The word स्त्र्याजीव (stryajeev) has a combination of four consonants. This is where things get interesting. Pani explains, "If you want to write a word like हृदरोग (hridrog). When you write with conventional systems, the way Unicode has been defined, because it is not included the Indian language grammar very well, the halant ends up acting as the conjunct maker. द (da) and र (ra) will join and it will end up becoming हृद्रोग (hridrrog). That becomes unacceptable. The southern and eastern languages are super strict about such incident vowel association. They will not even be able to read it anymore. In the Swalekh keyboard we introduced a yukta key on the top left. What we discovered

The Gboard layout in different languages